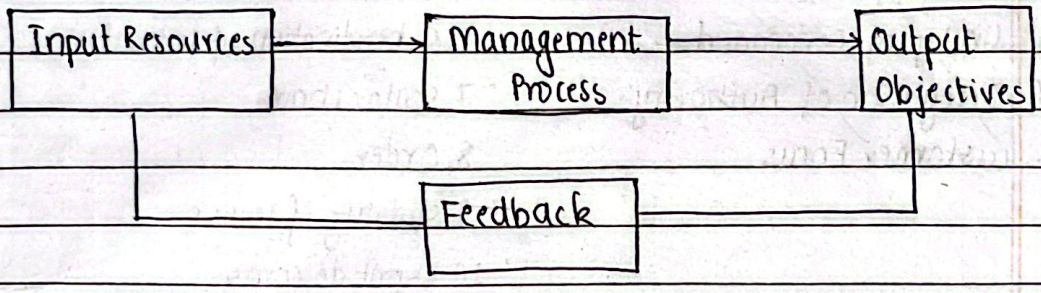
Unit - I

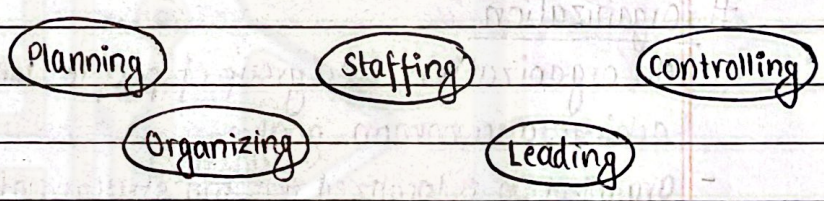
Management is the process of designing and maintaining an environment in which individuals, working together in groups, efficiently accomplish selected aims.



Types of Management : General Management, Functional Management, Public Administration, Non-profit management.

Functions of Management

1. Planning
2. Organizing
3. Staffing
4. Leading
5. Controlling



Level and scope of Management

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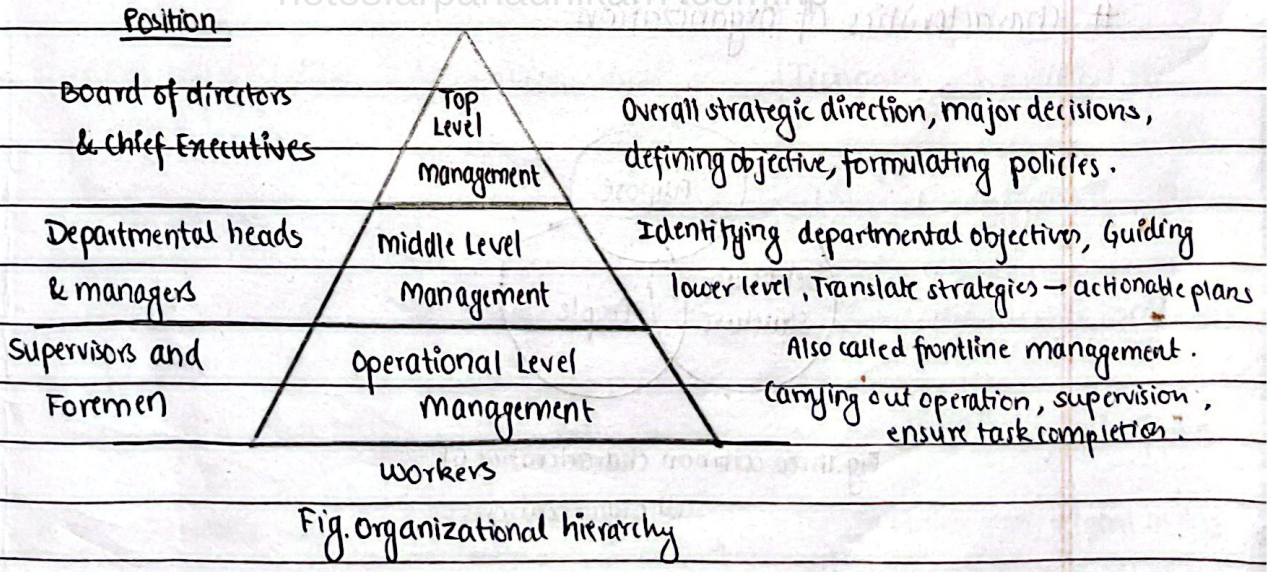

Fig. Organizational hierarchy

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Henri Fayol → Father of modern management theory
- Proposed 14 principles of management.

(DADU IS SO SECURE)

# Principles of Management	Fayol's Principles:
1. Management by objectives	1. Division of Work
2. Division of work	2. Authority & Responsibility
3. Substitution of Resources	3. Discipline
4. Span of control	4. Unity of command
5. Unity of Functions	5. Initiative
6. Unity of command	6. Subordination of interest
7. Delegation of Authority	7. Scalar chain
8. Customer Focus	8. Order
	9. Stability of Tenure
	10. Esprit de corps
	11. Centralization
	12. Unity of direction
	13. Remuneration 14. Equity

Organization

- An organization is a group of people who work together to achieve a common goal.
- Organization is localized ~~intential~~ deliberate structure of roles or problems.
- Organization made up of : People, Structure, Resources
- E.g. A company, a corporation, institution, union, association

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Characteristics of Organization

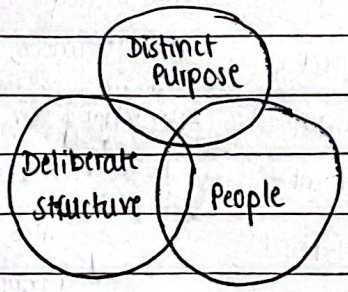

Fig. Three common characteristics of all organizations

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Golu DP Mein Cha Gaya

1. Strong Leadership
2. Specialization
3. Control
4. Flexibility
5. Organizational goals
6. Organizational structure
7. Teamwork
8. Division of work
9. Centralization
10. learn about customer Needs

1. Goal Oriented
2. Dynamic (keeps changing).
3. Pervasive (exists everywhere)
4. Multidimensional (involves multiple aspects of people, structure, tech. Not just abt hierarchy ;→ communication, relationship, process)
5. ~~Continuous~~ (~~ongoing process~~)
5. ~~Continuous~~ Control (Organization controls behavior of employee).
6. Group Activity. (Division of work).

Types of Organization

- Formal Org.
- Informal Org.
- Virtual organization

	Formal Org.	Informal Org.	Virtual organization
Formation	Officially created by management	Naturally formed	created through digital networks
Structure	Clearly defined hierarchy	No fixed structure	Flexible network structure
Communication	Official channel	Grapevine (informal)	Online communication tools
Rules	Written rules & policies	Unwritten norms	Technology based policies
Authority	Clearly defined	No official authority	Shared or contractual authority
Focus	Organizational goals	Social relationship	Collaboration across locations
Physical Presence	Yes	Within formal structure	No common physical office
Example	A company with clearly defined organizational chart where employee has specific job title & reporting structure	A group of colleagues who regularly have lunch together & share idea outside of formal work roles.	A team of designers located in diff. cities who collaborate on project using online platforms

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Engineering Management

- Engineering management is a field that combines engineering, business and management principles to lead & oversee engineering projects.
- Engineer managers are responsible for successful completion of project, operations & for ensuring industries run smoothly.

Importance of management in technology driven environments.

- Management is crucial for guiding adopting, integration & optimization of new technologies.
- Help to navigate through complexity of technology, aligning technology to business goals.

Key Points:
1. Data driven decision making
2. Innovation & adaptation
3. Collaboration & communication
4. Change Management
5. Cybersecurity
6. Alignment with business objectives
7. Technology Forecasting
8. Talent Management
9. Risk Management

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Engineering Functions in organization

1. Product Development
(Software development Roles: software developer, UI/UX designer, Project Manager, QA Engineer, Testes)
2. Operations
(coordinating teams & departments to improve product. Collecting feedback. Quality Assurance, cross-functional collaboration, Process optimization, Roles: Product Operation Manager, Product Operation Analyst)

data, metrics, performance.

Strategy, workflow, people

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3. IT systems (They do: Process data, Communicate, manage, secure E.g. Transaction Processing System (TPS), Office Automation System (OAS), Decision Support System, Executive Support System, Management Information System (sales report, convert data → report for managers)

4. Quality Assurance (Ensure software is reliable & fulfill needs, SQA activities: Testing, Functional Testing, Defect-tracking, Multi-testing, Continuous Integration & Development, Maintenance)

5. Others: (Engineering management, continuous Improvement, Compliance with regulations, Innovation, Mentorship & Development)

Roles & Responsibilities of Engineering Managers

1. Team Management
2. Project Oversight
3. Technical Guidance
4. Compliance
5. Risk Management
6. Stakeholder Communication
7. Process Improvement
8. Budgeting
9. Innovation
10. Talent Management
11. Quality control

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Unit-II

Planning

Planning is the process of setting goals, defining actions to achieve those goals, and organizing the resources and efforts needed to carry out those actions.

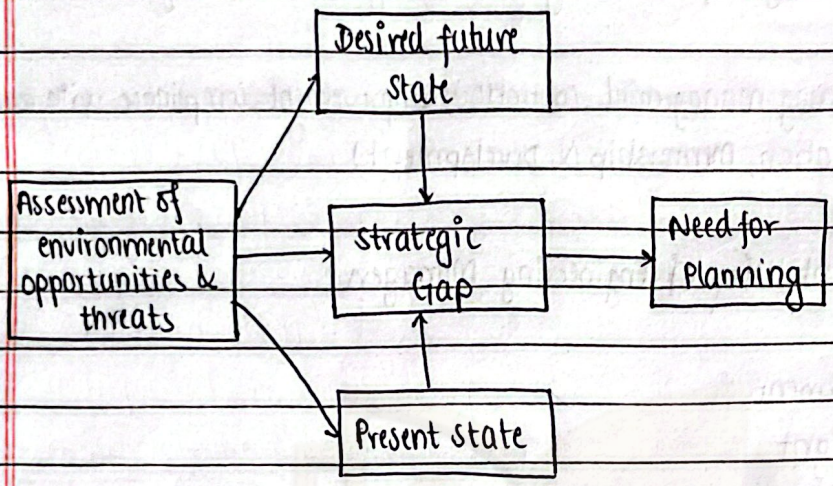
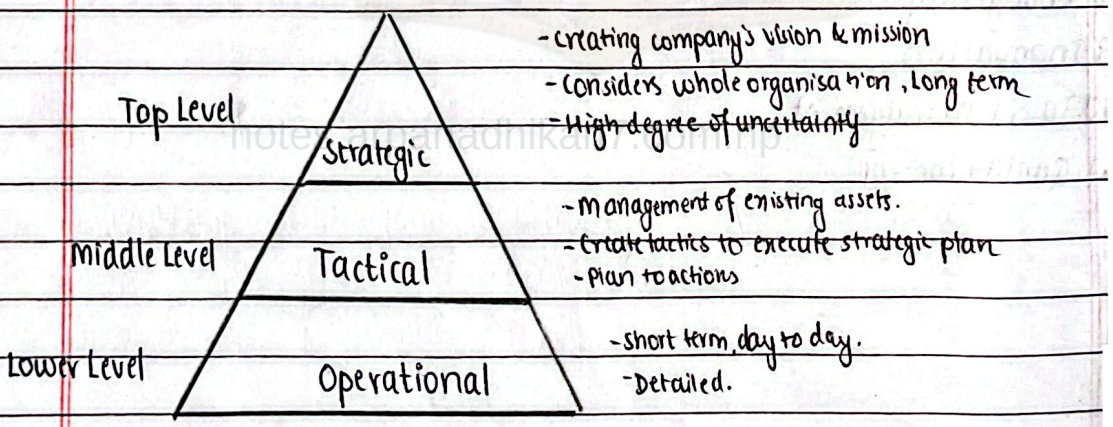


Fig. A Planning Model

Levels of Planning



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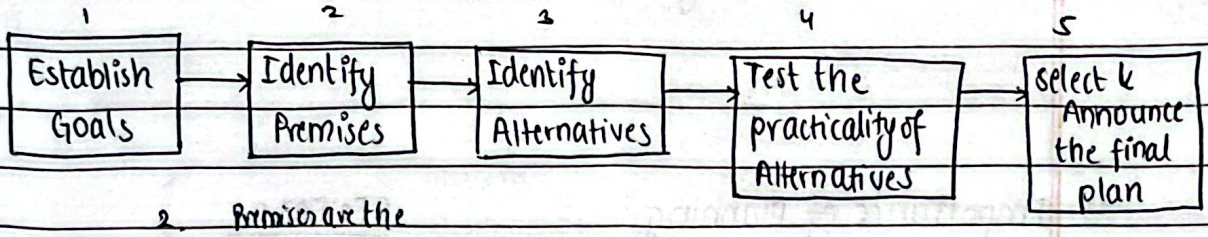
#steps in planning

Five steps:



2. Premises are the assumptions about future on which plans are based. E.g. Market conditions, resources, policies, technology.

4. Each alternative is evaluated based on factors like cost, feasibility, risks, benefits, available resources

5. When plan selected, plan is clearly communicated to all employees so that each understand their roles & responsibilities.

Tools for Planning (Planning tools help organize task, manage projects & allocate resources.)

(i) Strategic planning tools (Long term Planning & Overall direction)

1. SWOT Analysis (Strength, Weakness, Opportunities, Threats)
2. SMART Goals (Specific, Measurable, Achievable, Relevant, Time-bound)
3. PESTEL Analysis (Political, Economic, Social, Technological, Environmental, Legal)
4. Gap Analysis (Gap of current v/s desired position)
5. Vision & Mission Statement

(ii) Project ~~Management~~ Planning & scheduling Tools (Tools to plan & control project work)

1. CPM (Critical Path Method) {Identify activities which take longest time, proper scheduling}
2. Gantt Chart (visual timeline of project
3. PERT (Program Evaluation & Review Technique) {Helps to calculate project timing. Used in uncertainty timing of projects).

(iii) Digital Planning Tool (Using software for planning)

1. Project Management software
   (E.g. ClickUp, Trello, Microsoft Project

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(iv) Analytic & Data Driven Tools (making decisions based on data & feedback)

1. Surveys
2. Feedback Forms
3. Break-Even Analysis

Importance of Planning

1. Goal Clarity
2. Efficient Resource Use
3. Risk Reduction
4. Improved Decision Making
5. Coordination & control
6. Gives Right Direction
7. Motivates the Personnel

GENERIC

1. Goal Clarity
2. Efficient use of resources
3. Navigates Uncertainty
4. Encourage Creativity & Innovation
5. Risk Reduction
6. Improved decision Making
7. Coordination & control

Organizing

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Organizing in management is the process of (structuring team by) assigning tasks & resources to achieve a company's goals. ~~It also involves~~ delegating authorities, defining roles & coordinating activities

Organizing in management is the process of structuring a team by assigning roles, delegating authorities, allocating resources and coordinating activities to achieve company's goal.

Importance of Organizing : Efficiency, communication, Accountability

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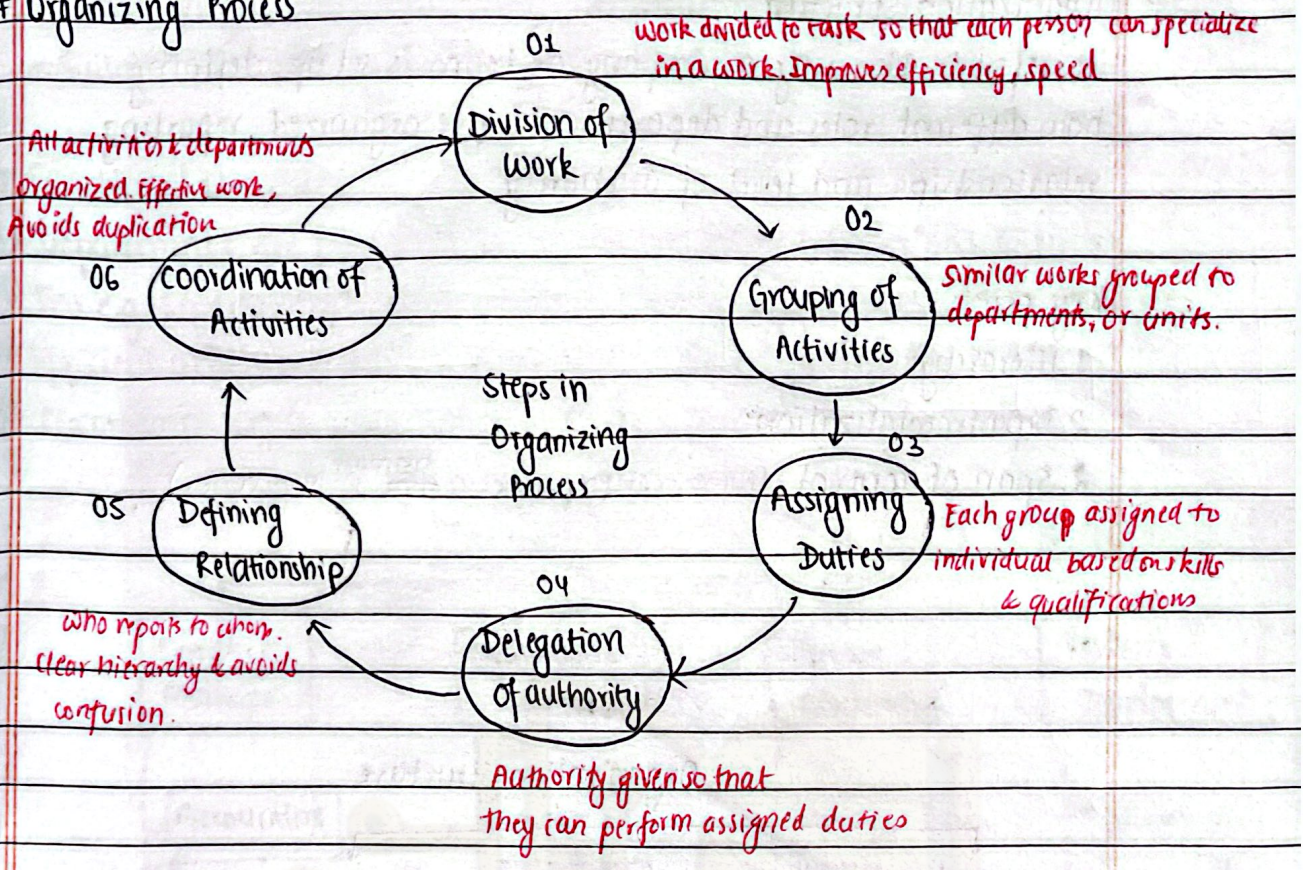
Organizing Process



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Organization structure

It refers to the way a company or team is set up, defining how different roles and departments are organized, reporting relationships and level of authority.

Key points:

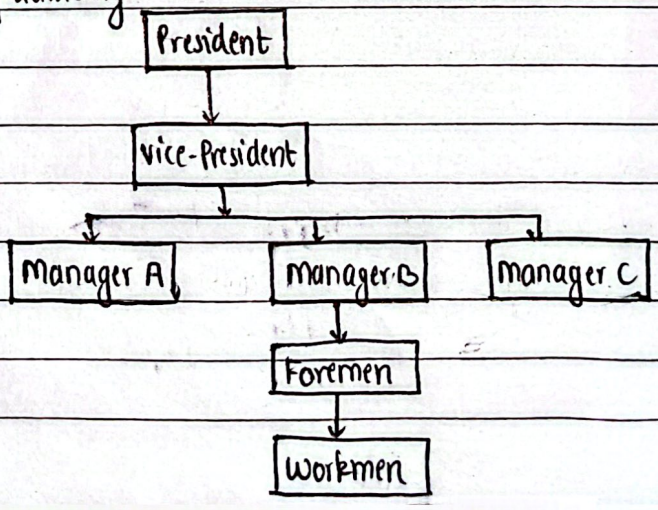
1. Hierarchy
2. Departmentalization
3. Span of control (Number of employees a manager supervises)

Organization structure

Traditional structure	modern structure
- Line	- Matrix
- Functional	- Network
	- Hybrid

1. Line structure

- Also known as scalar or military organisation, simple & oldest
- clear & direct chain of command
- Limited delegation of authority



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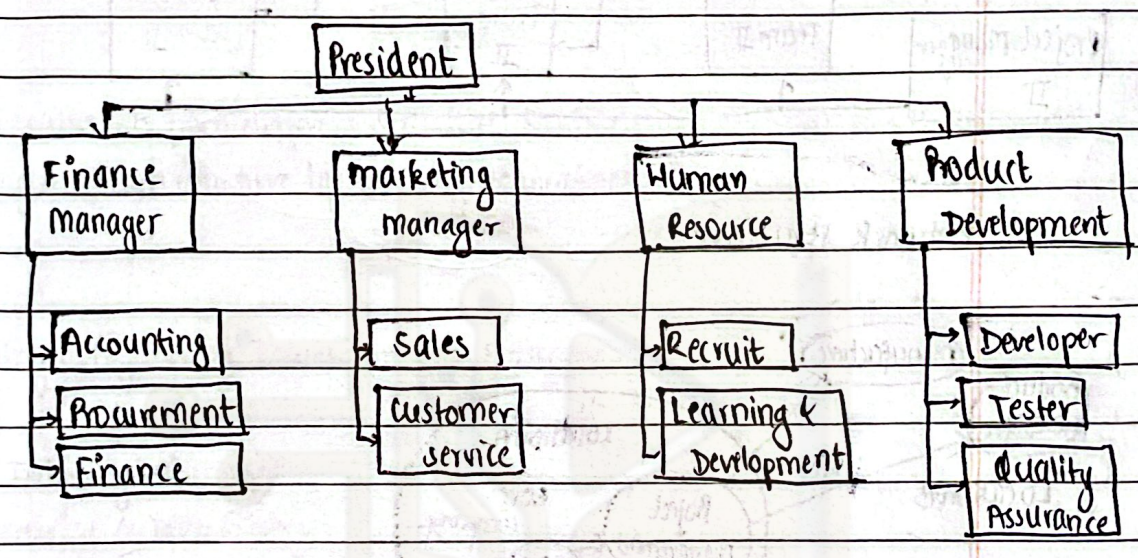
- Each employee has single supervisor. Decision making centralised
- Suitable for small organisations. can Lead to delay indecision making

2. Functional structure

Q- Departments are organised based on specialized functions & tasks.
- For e.g. separate dept for marketing, finance, operations, human resources.
- Efficient utilisation of skills. clear path in each department.
✓- Communication & collaboration betⁿ departments may be limited.



Modern structure

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1. Matrix structure
- Hybrid grid structure of project structure & functional structure.
- Involves dual reporting lines where employees have both functional managers & project managers.
- Functional manager → oversee employee's functional responsibilities.
  Project manager manages involvement in specific projects.
- Promotes sharing of expertise and knowledge across projects & functional areas
- May be complex to manage, conflicts of priority betⁿ two managers by employee

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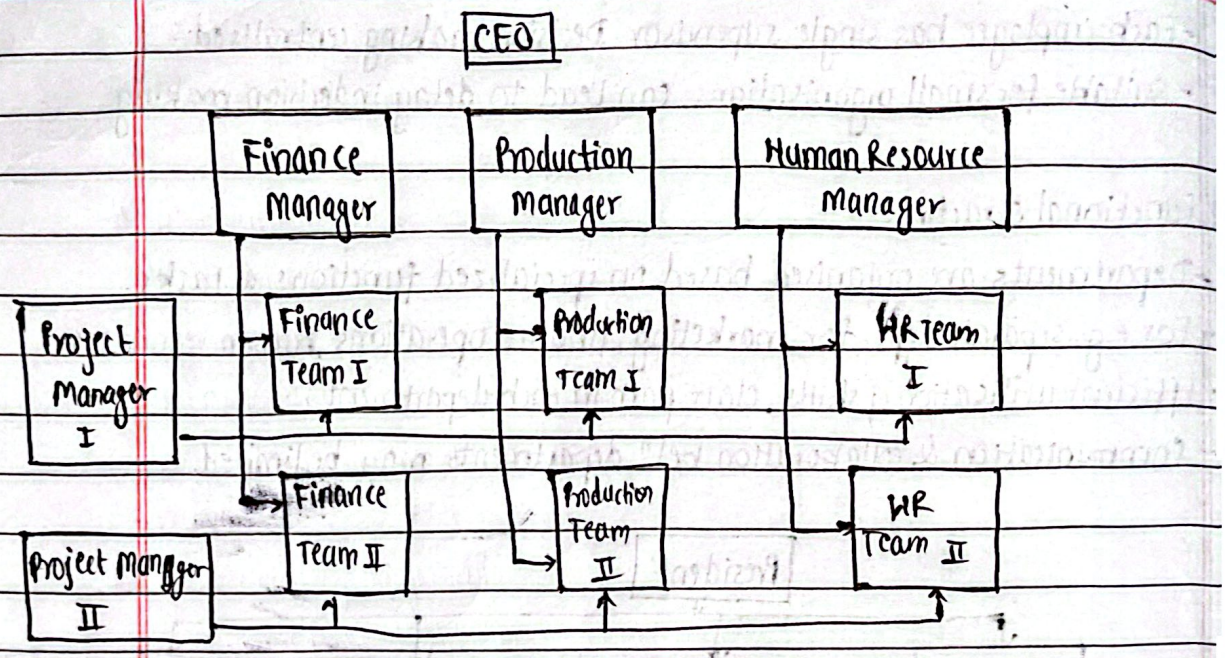
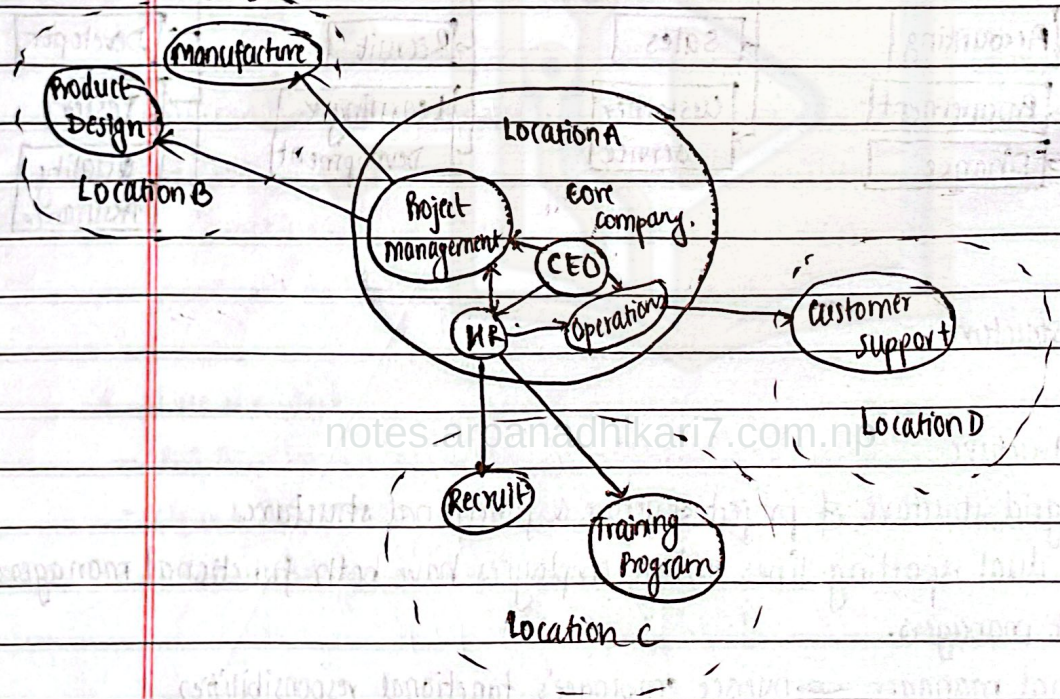


2. Network structure



- Decentralized business structure, where employee work in team to achieve goals.
- Connect with external org. (partner, freelancers) to perform activities instead of doing everything internally.
- Decentralized control / flexible structure / Tech-based communication
- main company focus on core task & outsources other task.

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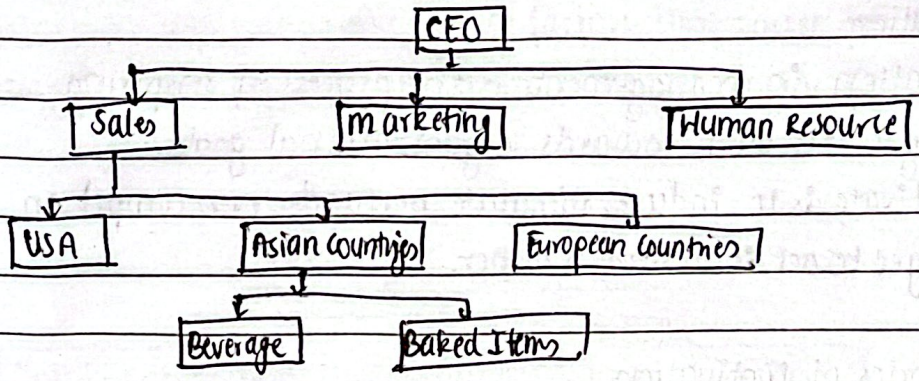
E.g. Software company. software design in-house, outsourced: coding, testing, marketing.

3. Hybrid structure



- Combines elements from different structure. Flexible structure that takes advantage of strength of each structure. try to mitigate weakness.

Planning & Organizing Issues for ICT Enterprises.

1. Rapid Technological change
2. Cybersecurity & Data Privacy
3. Agile work Environments
4. Talent management & skill Gaps
5. Globalization & virtual Workforce (Global Complexity).
6. Sustainability & Environmental Impacts
7. Regulatory compliance & Ethics
8. Scalability & Cloud Infrastructures
9. Customer Centric Planning
10. Data driven decision making.

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Types of Motivation:
1. Intrinsic Motivation (Personal fulfillment, satisfaction)
2. Extrinsic Motivation (done to obtain external rewards)

Chapter-3

Motivation & Leadership

Motivation

- Motivation in management is the process of inspiring employees to work towards organizational goals.
- To motivate is to induce, stimulate, persuade, even compel an employee to act in desired manner.

Theories of Motivation

Theories of motivation or motivation theory are the study of understanding what drives a person to work for achieving a particular goal or outcome.

1. Maslow's theory of Hierarchial Needs

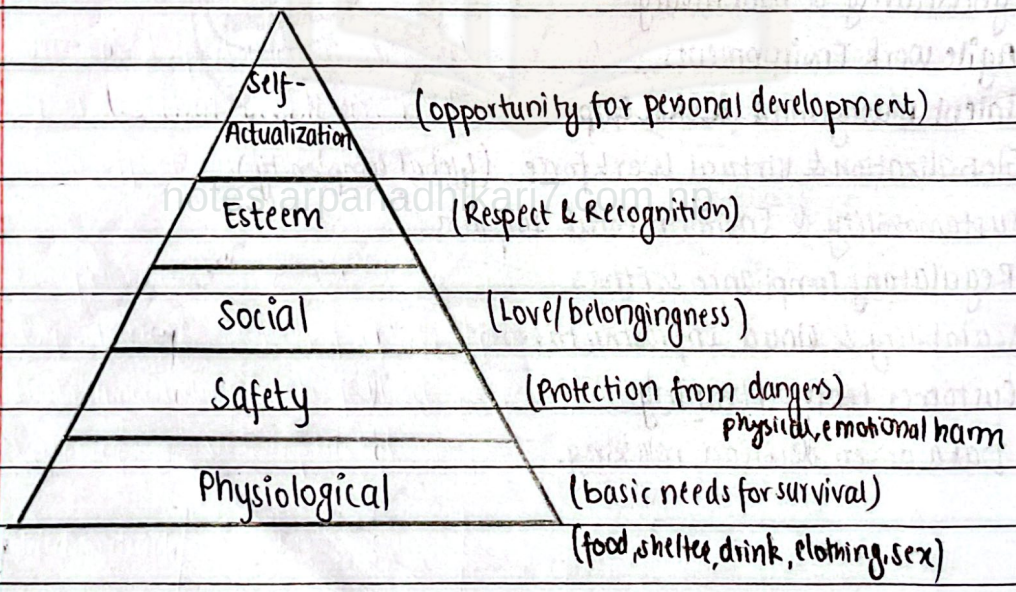

Fig. Maslow's Hierarchial Needs

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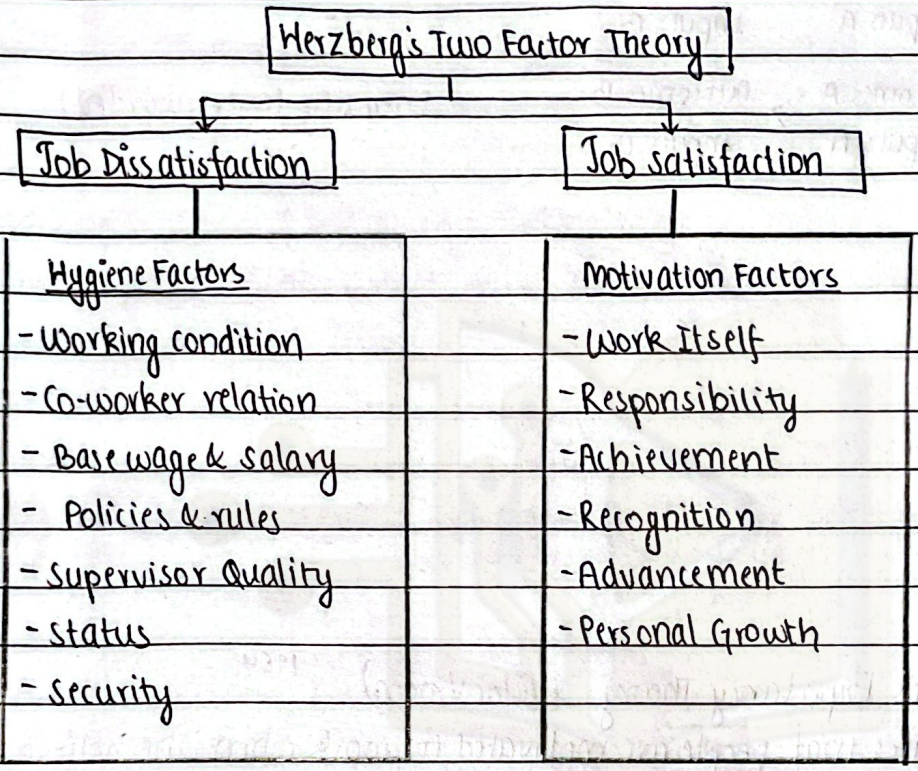
2. Herzberg's Two Factor Theory

- Dual Factor theory
- It states that there are certain factors in workplace that cause job satisfaction while separate set of factors that cause job dissatisfaction.
- Theory developed by collecting data from 203 engineers & accountants



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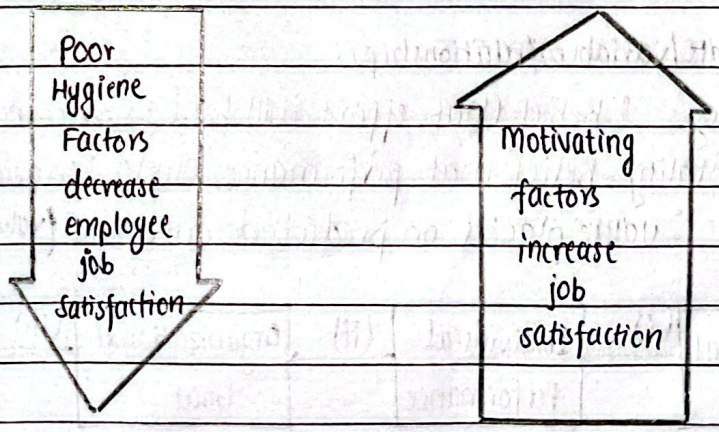

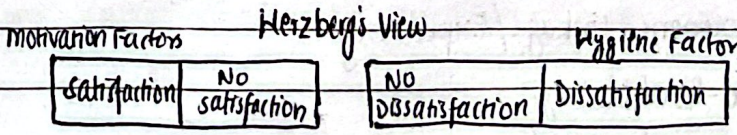

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3. Equity Theory 1963

Perceived Ratio Comparison			Employee's Assessment
$\frac{\text{Outcomes A}}{\text{Inputs A}}$	$<$	$\frac{\text{Outcomes B}}{\text{Inputs B}}$	Inequity (Underrewarded)
$\frac{\text{Outcomes A}}{\text{Inputs A}}$	$=$	$\frac{\text{Outcomes B}}{\text{Inputs B}}$	Equity
$\frac{\text{Outcomes A}}{\text{Inputs A}}$	$>$	$\frac{\text{Outcomes B}}{\text{Inputs B}}$	Inequity (Overrewarded)

4. Vroom's Expectancy Theory (Victor Vroom) 1964

It states that people are motivated to work when the believe their efforts will lead to positive outcome and rewards.

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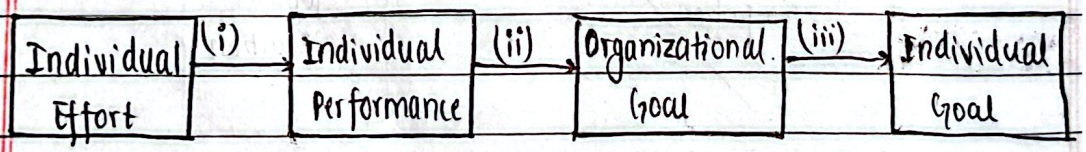
Components/variables/relationships:

(i) Expectancy : belief that effort will lead to performance
(ii) Instrumentality : belief that performance will lead to predicted outcome
(iii) Valence : value placed on predicted outcome (value of reward for person)



(i) Effort-Performance Linkage/Expectancy
(ii) Performance-Goal Linkage
(iii) Attractiveness of award

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Techniques of Motivation

1. Provide incentives & rewards
2. Encourage healthy work-life balance
3. Provide opportunity for personal growth
4. Provide responsibility
5. Provide meaningful work (work itself) {Interesting job, engaging, less repetitive}
6. Provide positive feedback
7. Encourage teamwork
8. Provide clear communication

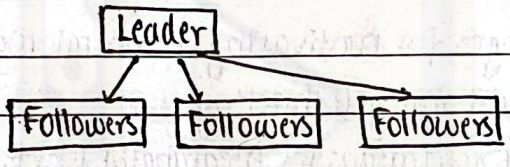
Leadership

Leadership is the act of influencing & guiding others.
Leadership style refers to how a leader guides, motivates & manages others.

Styles:

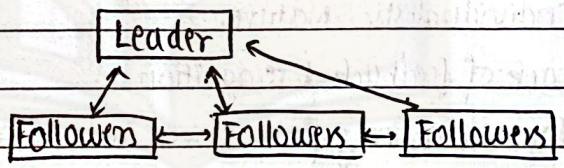
1. Autocratic

Leaders make decision independently
Quick decision



2. Democratic

- Participative Leadership
- Require discussion/debate for decision



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3. Servant Leadership

- Prioritize of followers needs and views.
- ~~Instead of concentrating on results~~, Empathy & support, Encourage personal growth of employees, build trust & strong relationships. Adv: High satisfaction, better morale. , slow decision.

4. Transformational Leadership

- Leaders are highly inspirational, motivational, passionate & energized. Encourage follower to exceed their own self. Employee growth.

Risk of burnout, High expectations may burnout employees.

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Learning Organization

Organization that
- Facilitates learning among team members.
- Constantly transform itself (adaptability)
- Can adapt to challenges in its environment

Characters:
1. Continuous Learning
2. Knowledge Sharing
3. Open communication
4. Adaptability
5. Innovation & Experimentation
6. collaborative Environment
7. Shared Vision & Goals
8. Leadership that supports learning
9. Employee Empowerment
10. Reflective Practices

Challenges for motivating technical workforce
1. Rapidly changing technology
2. High Expectations for Meaningful Work
3. Individualistic Nature
4. Lack of feedback & recognition
5. Communication gap
6. Lack of autonomy & empowerment
7. Lack of work-life balance.
8. Lack of growth & development
9. Deadlines & pressure.

Strategy:
1. Provide challenging & meaningful problems
2. Encourage Autonomy (less micromanagement)
3. Create learning driven environment
4. Recognize contributions, not only results
5. Provide flexible time
6. Brainstorming sessions
7. Using modern tools & technologies
8. Align work with purpose (explain why project matters)
9. Address burnout & stress

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Unit-4 Human Resource Management & Control

Human Resource Management (HRM) is the practice of recruiting, hiring, deploying and managing organization's employees.

#Functions of Human Resource Management

HRM serves as linchpin between organization's objectives & its workforce.

1. Talent Acquisition & Recruitment
2. Employee Development & Training
3. ~~Perfo~~ Performance Management & Appraisal
4. Employee Relations & Engagement
5. Compliance with Employment Laws & Regulations
6. Organizational Development & change management
7. Career Development
8. Maintaining safe & healthy Environment

#Job analysis is the systematic process of collecting and analyzing information about a job to understand its duties, responsibilities, req^{d} skills & working conditions. Foundation for job specification & job description.

- Updation of job roles by analysis.

#Job description is a written statement that clearly explains duties, responsibilities and scope of job.

Elements: job title, duties & responsibilities, reporting relationship, working conditions, performance standards

#Job specification is a detailed description of qualifications, skills, experience and personal attributes required to perform a job effectively.

Elements: Educational qualifications, experience, technical skills, soft skills, Physical/mental abilities

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Recruitment & selection

1. Job Analysis
2. Planning & Forecasting
3. Job posting & Advertising
4. Candidate Sourcing
5. Initial Screening
6. Preliminary Interviews
7. Employment Testing
8. In-depth Interviews
9. Reference Checks (Bg check)
10. selection & Job offers
11. Onboarding

Human Resource Training

On the Job Training (OJT)

1. At workplace.

Methods: 1. Job Rotation
2. Apprenticeship Programs
3. Coaching / Mentoring (by senior employee)
4. Internship

Off the Job Training (Off-JT)

Away from workspace or job roles.

Methods:
1. Computer Based Training (modules)
2. workshop / seminars
3. Sandwich courses
4. Case-studies
5. simulation / Role-play
6. Games / Experential Learning

Performance appraisal is a process that evaluates an employee's job performance & contributions to company.

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Performance appraisal methods:

(i) Traditional method
1. Ranking method
2. Grading Method
3. Paired comparison method
4. Essay method (supervisor writes)
5. checklist Method
6. Graphic Rating scale method (rated on traits like Attendance, teamwork, quality of work etc.)

(ii) modern method
1. management by objectives (MBO) (Employee & manager set measurable goals. performance evaluated based on goal achievement)
2. 360 Degree Feedback (Feedback from superior, subordinates, peers, customers)
3. Assessment center Method (Employee participate in discussion, role play, case studies.)

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#challenge in managing people of ICT workforce.

1. Rapid technological change
2. maintaining employee engagement in fast paced environment.
3. work-life balance
4. Managing Remote work
5. Change Management.
6. Addressing potential burnout from pressure
7. constant Adaption
8. ~~Eff~~ cybersecurity concerns.

Control

Control is a managerial function that involves monitoring, comparing & correcting performance of employees and activities to ensure organizational goals are achieved as planned.

Importance:
1. Achieving Organizational goals
2. Identifying Deviations
3. Maintains Discipline
4. Improves Efficiency (minimize waste of resources)
5. Improves Coordination
6. Facilitates better decision making
7. Adaptation to changes.
8. Identifies training needs
9. Motivates Employees
10. Improves Employee Performance

Control Process:
1. Establishing Standards
2. Measuring Performance
3. Comparing Performance With standards
4. Identifying cause of deviations (if exist)
5. Take Corrective Action

Types of Control:

1. Preventive Control	2. Detective Control	3. Corrective Control
Avoid problems before they occur.	Identify problems, detect problems.	Fix problems after detected.
Time Before	Before/After	After

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Techniques for Control

Control technique help ensure work is done correctly, efficiently.

1. Budgetary control (Planning & controlling financial resources)
2. Quality control (Ensuring product meet standards)
3. Performance Appraisal (Assessment of employee & system)
4. Risk Management (Identifying, analysing, reducing potential risks)
5. Automation & AI monitoring (using automated tools & AI to monitor & control process)
6. Benchmarking (comparing performance with best performing companies or standards)
7. Statistical control (Use statistical data to monitor performance & quality)

ICT Tools

1. Project Management Tools (ClickUp, Trello, Microsoft Project)
   Track tasks, deadline, team progress, identify delays early.
2. Communication & collaboration Tools
   E.g. Meet, Zoom, Teams
   Enable seamless team communication
3. Quality Assurance & Testing tools
   E.g. Selenium, TestRail
4. Version control system
   E.g. Git, Github, Bitbucket
5. Cybersecurity & compliance Tools
   E.g. Firewalls, Antivirus software
6. Cloud Computing Tools
   E.g. Google drive, AWS
7. Data Analytics TOOLS (Analyze project data & generate reports).
   E.g. Tableau

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